

C1 IP internal structure — conv_c1_int8_fixedpoint (8-PE systolic MAC array)

Solution conv_c1_int8_fixedpoint_proj/solution1 • xc7z020clg400-1 (PYNQ-Z2) • 10 ns clock • 28×28×1 → 5×5 conv, 'same' pad → 28×28×6, int8 × int8 → int32

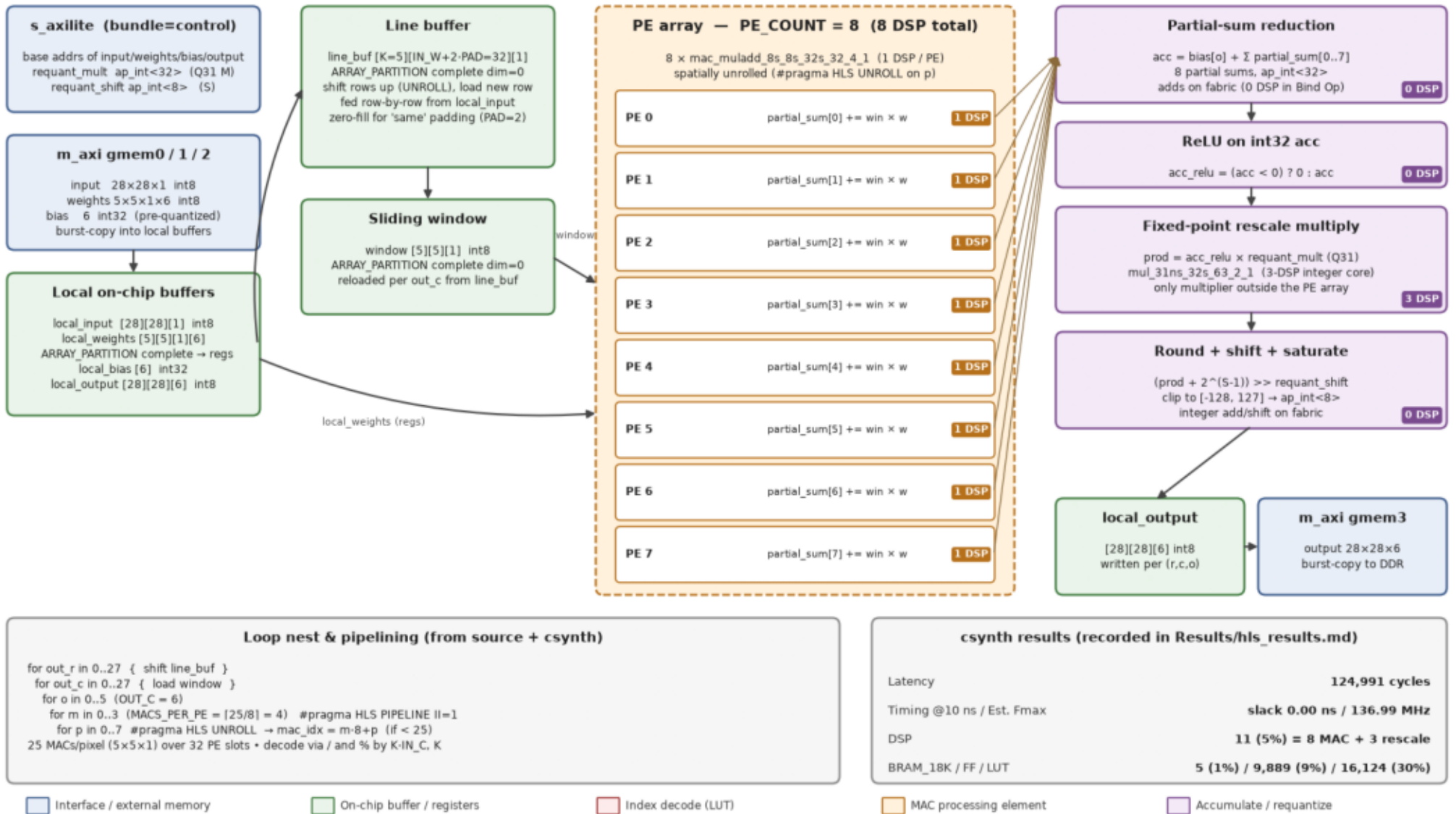


Figure 1 — C1 (conv_c1_int8_fixedpoint). PE_COUNT = 8 spatially-unrolled int8 MAC PEs (8 × mac_muladd_8s_8s_32s_32_4_1, 1 DSP each) fed from a fully partitioned 5×32 line buffer and 5×5 window; fixed-point requantize via one mul_31ns_32s_63_2_1 (3 DSP). Totals: DSP 11 (5%), BRAM_18K 5 (1%), FF 9,889 (9%), LUT 16,124 (30%); latency 124,991 cycles; slack 0.00 ns at 10 ns, est. Fmax 136.99 MHz; xc7z020clg400-1.

C3 IP internal structure — conv_c3_int8_fixedpoint (6-PE partial-sum split + LUT index decode)

Solution conv_c3_int8_fixedpoint_proj/solution1 • xc7z020clg400-1 (PYNQ-Z2) • 10 ns clock • $14 \times 14 \times 6 \rightarrow 5 \times 5$ conv, 'valid' $\rightarrow 10 \times 10 \times 16$, $\text{int8} \times \text{int8} \rightarrow \text{int32}$

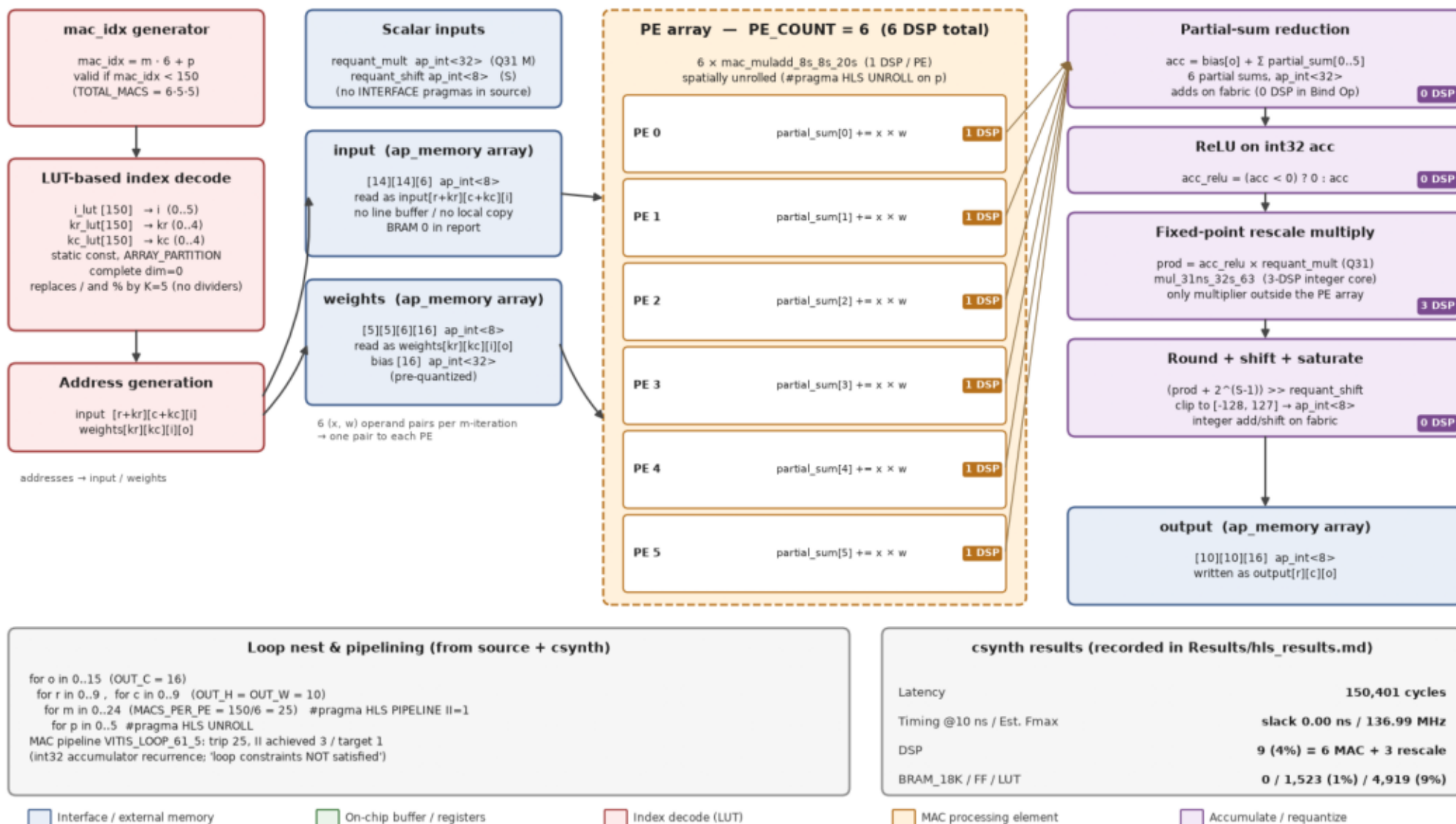


Figure 2 — C3 (conv_c3_int8_fixedpoint). PE_COUNT = 6 partial-sum split (6 × mac_muladd_8s_8s_20s, 1 DSP each) with division-free (i, kr, kc) decode from three 150-entry fully partitioned LUTs; fixed-point requantize via one mul_31ns_32s_63 (3 DSP). MAC loop VITIS_LOOP_61_5: trip 25, II 3 (target 1). Totals: DSP 9 (4%), BRAM 0, FF 1,523 (1%), LUT 4,919 (9%); latency 150,401 cycles; slack 0.00 ns, est. Fmax 136.99 MHz.